

Curriculum Vitae

Matthew N. Cook

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[Google Scholar](#)

Matt Cook is a multidisciplinary technology consultant, in-demand speaker, and cited scholar. He is a proven expert across the entire 3D data lifecycle; from scanning (e.g. photogrammetry), to immersive (AR/VR) analytics, to rapid prototyping and 3D printing. Matt has published R&D in multiple fields, including Architecture, Chemistry, Information Science, and a range of Humanities. Currently he is interested in automated workflows that combine the use of handwriting transcription technologies, genAI, and the Metaverse.

Education

- M.L.I.S., University of Oklahoma. May, 2019. Advisor: Kun Lu, PhD.
- M.A., Philosophy. Thesis: "Extended Perception: The Scope and Limits of Cognition." Advisor: Jim Hawthorne, PhD. University of Oklahoma, May 2012.
- B.A., Philosophy, *cum laude*. Advisor: Gene Witmer, PhD. University of Florida, May 2010.

Areas of Specialization

Digital Scholarship/3D Data/Philosophy of Mind

Professional Experience

- Digital Scholarship Program Manager, Harvard Library (Summer 2019-Present)
 - New unit and position dedicated to cross-campus emerging technology consulting.
 - Co-Developed multiple technology spaces and University-wide services
 - Leads interdepartmental project teams across organization (3D, AI, VR, etc.).
- Head of Emerging Technologies, University of Oklahoma Libraries (late 2017-Summer 2019)
 - Expanded team to 5 FTEs, including full-time software developer.
 - Expanded student staff to 20+ undergraduates across 3 campus locations.
 - Membership on University Libraries' senior (executive) committee
- Emerging Technologies Coordinator, University of Oklahoma Libraries (2016-2017)
 - Implemented and managed campus-wide system of Makerspaces/Visualization Labs
 - Developed and deployed custom VR workstations used across 12 academic disciplines.
 - CO-PI on IMLS National Leadership Grant on 3D/VR technology implementation
- Emerging Technologies Librarian, University of Oklahoma Libraries (2013-16)
 - First such position at University of Oklahoma Libraries
 - 2015 Library Journal "Mover and Shaker"
 - Co-Coordinated Learning Commons opening (Helmerich Collaborative Learning Center)

Major Projects & Services

- Custom Systems & Services
 - Sparq Labyrinth – *design development/prototyping/testing/peer-reviewed publication*
Interactive mindfulness tool to counteract stress and promote workplace mindfulness.

- Deployments: U of Oklahoma (Norman and Tulsa campuses); UMass, Amherst; Northwestern Oklahoma State University; Concordia University, Nebraska; CUNY.
- OU Libraries NavApp – *vendor negotiation/testing/front-end design/deployment/marketing* Bluetooth beacon-based smartphone for indoor guidance to campus resources:
 - Vendor liaison (Aruba, HP Enterprise, RFIP) + \$100,000+ CFP and negotiation.
 - Scaled to ~5000 active campus users, 2016 Campus Technology's *Innovator Award*
- Oklahoma Virtual Academic Laboratory (OVAL)– *Hardware design/software architecture/usability testing/course integration/ marketing/instructional design* Discipline agnostic system for the analysis and manipulation of 3D scholarly assets
 - 20+ unique (for credit) course integrations. 1000+ students supported, per year.
 - Implementations at: Arizona, North Texas, Stanford, Miami, UC Santa Cruz and more.
- Harvard Library specific:
 - 3D Scanning Service – *production/preservation/metadata schema/client relations*
 - Integrations across Comparative Languages and Literatures; Earth and Planetary Sciences; Humanities; History; Gen Ed; and History of Art and Architecture.
 - Immersive Showcase: Virtual teaching and learning environment (expected Fall 2025)
 - Virtual Harvard: Administered (Matterport) scans for Harvard College Libraries.
 - 3D Printing: Developed and supported campus-wide 3D printing service
 - Longhand: Immersive “Word Cloud” text visualization tool.
- “Innovation Space” consulting (visualization labs, makerspaces, etc.)
 - Innovation @ the EDGE makerspace (founder)
 - Flexible experimentation space featuring electronics, XR, 3D printing, etc.
 - Winner: Campus Technology *Impact Award*, 2017 (for Innovation @ the EDGE)
 - Tom Love Innovation Hub *VisLab* (founder)
 - Couch dormitory (U of Oklahoma), dormitory-based 3D Printing lab (co-founder)
 - Consulting technologist:
 - University of Oklahoma School of Law; Harvard Business School *Innovation Lab* (AR/VR Studio; Harvard Medical School (Countway Library makerspace); Harvard Graduate School of Design, *Digital Research Studio*; HUGSE *Data Studio*.

Grants, Awards, and Service

- Co-PI. 3D Research Data Curation Framework (*3DFrame*). IMLS NLG, ~\$600k
- Instructor-of-record, Harvard College, Junior Tutorial 98R (“Future Human”)
- “Mover and Shaker,” Library Journal’s ‘Movers and Shakers’ (March 2015)
- Innovator’s & Technology Impact Awards. Campus Tech Magazine (2016, 2017)
- Co-PI. *Lib3D/VR* IMLS National Leadership Grant (~\$95k, FY 2017)
- University of Oklahoma Libraries *Innovator’s Award* (2013 & 2017)
- Oculus *NextGen* - faculty member (2018)
- Technical lead. Dean’s Fund/Milton Fund, *Shund Collective* (internal) grant awards, \$60k (FY22)
- Motsepe Presidential Research Accelerator Fund for Africa (~\$100k), senior technologist (2023)
- Harvard University’s *Elson Fund* Grant (\$5k). Technical lead
- Ed Cline Faculty Development Award. Technical Lead
- OU College of Arts and Sciences travel grant & OU Philosophy travel grant (2011).
- Member, American Library Association and Oklahoma Library Association, 2015-present
- Referee: *Library Hi-Tech*, *Virtual Reality*, *Information and Learning Sciences*, *Journal of the Medical Library Association*, *Preservation*, *Digital Technology & Culture*, *Journal of Academic Librarianship*, *Library Quarterly*

- IVY+ Digital Scholarship working group (Director-level library cohort across Ivy+ institutions)
- Member, IIF (International Image Interoperability Framework) 3D sub-group
- IDEA (Innovation, Disruption, Enquiry, Access) Institute on AI – Cohort #1 Fellow. Summer 2021
- Digital Scholarship Support Group (DSSG) standing committee member. Harvard University
- Advancing Open Knowledge grant, technical support:
 - *Enhancing Slavery, Abolition, and Emancipation* (2020), *3D Printing at Harvard Library* (22)
- Born Digital Stewardship Advisory Group – Harvard Library, 2022-24
- The Carpentries, certified instructor
- DS Discovery Program (UC Berkely), Project Partner for Longhand data visualization tool
- Harvard HILT *Teaching and Learning Consortium* (TLC), 2022
- Instructor-of-Record for Special Concentration Tutorial, Harvard College, Fall 2023
- Digital Humanities Summer Institute (“Out of the Box Text Analysis”), onsite completion
- IIF 3D Technical Specification Group (TSG), Interoperability working meeting. January 2024
- Leadership Strategies for the Individual Contributor leadership development program, Harvard Center for Workplace Development, 2025.
- Digital Scholarship Discussion Group, Harvard Library. Ex Officio Convener, 2024-present

Research

Peer-Reviewed:

- Zucker, D.M., Choi, J., Cook, M. & Croft, J. (2016). “Effect of labyrinth walking in an academic library.” *Journal of Library Administration*, 56(8).
- Cook, M. (2018). “Virtual Serendipity: Preserving Embodied Browsing Activity in the 21st Century Research Library.” *Journal of Academic Librarianship*, Volume 44, Issue 1, Pages 145-149.
- Cook, M., and Betsy Van der Veer Martens. “Managing Exploratory Units in Academic Libraries.” *Journal of Library Administration* (2019): 1-23.
- Hall, Nathan Frank, Juliet Hardesty, Zack Lischer-Katz, Jennifer Johnson, Matt Cook, Julie Griffin, Andrea Ogier et al. “Challenges and Directions in 3D and VR Data Curation.” (2019).
- Poher, Elizabeth, and Matt Cook. “Thinking in Virtual Spaces: Impacts of Virtual Reality on the Undergraduate Interior Design Process.” *International Journal of Virtual and Augmented Reality (IJVAR)* 3, no. 2 (2019): 23-40.
- Cook, M. Grime, J. “Motivations, Design, and Preliminary Testing for a 360° Vision Simulator.” *Virtual Reality: XR (VR, AR, MR) and Immersive Learning Environments* (special issue). 2019.
- Cook, Matt, Zack Lischer-Katz, Nathan Hall, Juliet Hardesty, Jennifer Johnson, Robert McDonald, and Tara Carlisle. “Challenges and strategies for educational virtual reality.” *Information Technology and Libraries* 38, no. 4 (2019).
- Qin, T., Cook, M. and Courtney, M., 2020. “Exploring Chemistry with Wireless, PC-Less Portable Virtual Reality Laboratories”. *Journal of Chemical Education*, 98(2), pp.521-529.
- Cook, M., 2024. “The lifecycle of 3D data in academic libraries: A survey of methods and implications for information professionals”. *The Journal of Academic Librarianship*, 50(5).

Panel/Editor Reviewed:

- Cook, M., & Croft, J. B. (2015). “Interactive mindfulness technology: A walking labyrinth in an academic library.” *College & Research Libraries News*, 76(6), 318-322.
- Cook, M., Endres, W., & Kerlinkus, W. (2016) “OVAL: A Virtual Ecosystem for Immersive Scholarship and Teaching.” *Digital Humanities 2016*, Krakow.

- Pober, E., & Cook, M. (2016). "The Design & Development of an Immersive Learning System for Spatial Analysis and Visual Cognition". 2016 Conference of the Design Communication Association, Bozeman MT, 2016.
- Cook, M., & Waller, J. (2018). "Banned Objects Week: A Pre-emptive defense of Intellectual Freedom in the 21st Century Research Library." ALA's Journal of Intellectual Freedom and Privacy.
- Lischer-Katz, Z., Cook, M., & Boulden, K. (2018) "Evaluating the Impact of a Virtual Reality Workstation in an Academic Library: Methodology and Preliminary Findings." The Annual Meeting of the Association for Information Science and Technology. Vancouver, BC.
- Cook M., and Lischer-Katz, Z. 2021. "Practical Steps to an Effective Virtual Reality Course Integration" *College & Undergraduate Libraries*, 27(2-4), pp.210-226.
- Lischer-Katz, Z. and Cook, M., 2022. Virtual Reality and the Academic Library of the Future. *Transactions of the American Philosophical Society*, 110(3), pp.185-210.

Book Chapters, Reviews, and Editorships:

- For Reference & User Services Quarterly (RUSQ): "Makerspaces Top Trailblazing Projects" (Spring 2015); "The Library Innovation Toolkit: Ideas, Strategies, and Programs" (Fall 2015); and Reference & User Services Quarterly. "Managing Creativity: The Innovative Research Library"
- Cook & Thompson. "Strategic Approaches to Immersive Learning." Special issue - Information and Learning Sciences, 122(7/8), pp.442-463.
- Cook, M.N., & Lischer-Katz, Z. (2019). Integrating 3D and VR into research and pedagogy in higher education. In K. Varnum (Ed.), *Beyond reality: Augmented, virtual, and mixed reality in the library*.
- Van Epps, A., Cook, M. and Berstler, S., 2022, March. Collaborative Implementation of a Semi-Automated 3D Printing Service. Rowman & Littlefield.

Presentations (select)

- "Extended Perception via Structured Space," October 2011, Long Island Philosophical Association.
- "Extended Perception: The Limits of Mentality," Central States Philosophical Association, Columbia Missouri (University of Missouri), September 2012.
- "Spatial Study Tools." Canadian Library Association Annual Conference and Trade-show, 2013.
- "Barriers and Solutions to Wellness Using Labyrinths" (with Donna Zucker) at Improving Public Health for Underserved Populations conference. Oklahoma State University. April, 2016. *
- "Helping students navigate the campus of the future." (with Diana Laboy-Rush) at the Association of College and University Technology Advancement (ACUTA) annual conference. San Diego, 2016.
- "OVAL: A Virtual Ecosystem for Immersive Scholarship and Teaching." Digital Humanities 2016.
- "Immersive Education Using Virtual Reality" (with James Adams, Dartmouth). Library Information Technology Association. Fort Worth, TX, Nov. 2016.
- "Immersive Scholarship: Virtual Reality for Research and Instruction." UofArizona, Feb 2017. *
- ["VR In the Trenches: Addressing the Preservation Challenges of Virtual Reality for Scholarship."](#) (With Zack Lischer-Katz). Coalition for Networked Information Spring 2017 Meeting. ABQ, NM.
- "Inventing an Academic Career." *Poetics of Invention* Dream Course. Oklahoma, September 2017.
- ["Virtual Reality at OU Libraries - An Overview for PennImmersive."](#) With Zack Lischer-Katz. 2017.*
- "The Collaborative College: Partners on the Cutting EDGE." (with Melissa Ski, Fred Jones Jr. Museum of Art). 2017 School Library Journal, Maker Workshop. Online. October 2017.
- "VR@OU: Immersive Visualization on Campus and Beyond" – OSU Digital Humanities Forum.
- "Virtual reality for design studio critiques. Experiences of students and impacts on the design process" (With Elizabeth Pober). Interior Designers Educational Council (IDEC). March 2018.

*Invited talk.

- “Evaluating Virtual Reality Use in Academic Library-Supported Course Integrations: Methodology and Initial Findings” (Poster, with Zack Lischer-Katz). ALISE ‘18 The Expanding LIS Education Universe, February 6-9, Denver, Colorado.
- “Expanding access to library collections in three dimensions” with Zack Lischer-Katz. The Past, Present, and Future of Libraries, Conference. American Phil Society, Philadelphia, PA (Sept. 27-29).
- “Developing Library Strategy for 3D & Virtual Reality Collections” with Robert McDonald (CU Boulder), and Julie Griffin (VTech). EDUCAUSE. Denver, Colorado (November 2, 2018).
- “Integrating VR into Pedagogical Goals.” University of California Santa Cruz Library, the Humanities Institute, and The Center for Innovations in Teaching and Learning. March, 2019.*
- “Building and Supporting AR and VR Makerspaces.” U Buffalo’s *Center for Tomorrow*. April, 2019.*
- “The Expanding Role of the Academic Librarian.” Oklahoma Chapter of the Association of College and Research Libraries. Northeast Campus of Tulsa Community College. November 2019.*
- “Assessing Innovation.” LLAMA Innovation Incubator (webinar). January, 2020.
- “Immersive Design.” iLRN 2020: Immersive Learning Research Network.
- “Seeing at Scale - Artificial Intelligence and Access to Manuscript Materials.” Panelist. Harvard University. April 2021.*
- “Realizing the Untapped Potential of VR/AR for Libraries.” RUSA. September 22, 2021. *
- “The Library of the Future at Harvard Library.” Guest on Microsoft Libraries and Museums Podcast. <https://anchor.fm/microsoft-librariesmuseum/episodes/The-Library-of-the-Future-at-Harvard-Library-e19qev0> *
- “The Lifecycle of 3D Data.” Florida Digital Humanities Conference, April 2022. Flagler College.
- “Virtual Reality in the Humanities.” University of Arizona, School of Information. October 2022.*
- “Longhand: Text Tokens as 3D Object Arrays in Virtual Reality”. Greater Boston Digital Research & Pedagogy Symposium. MIT. April, 2023.
- “The Lifecycle of 3D Data”. Princeton University Library. September 22, 2023.*
- “Who Curates the Machine: A Pilot Data Extraction Project for Digitized Tabular Content”. Cook, Fay, and Wones. Digital Library Federation (DLF) annual meeting. St. Louis, MO. 2023.
- “Immersive Worldbuilding 101”. Southern Utah University *XR Summit*. Plenary. March, 2024.*

Software Proficiencies

- Design:
 - Blender - Computer-aided Design (CAD) software.
 - Adobe cs6 – Photoshop + Premier Pro + Illustrator
 - Autodesk Suite - 3Ds Max; Recap (photogrammetry); Revit, etc.
- Development:
 - Python – OCR/HTR, NLP, computer vision, data management, web scraping
 - Pandas, HandPrint, SpaCy, Gensim, OpenCV, Blender Python, Selenium
 - Unity3D – immersive application development
 - Git, AWS, Digital Ocean, Jupyter, Google Cloud Translate
- Web Publishing:
 - Scalar, Omeka, WordPress, Drupal, Open Science Framework, Dataverse
 - Sketchfab - 3D repository (username: Cook4986)
 - Domain of One’s Own - medium-term hosting (campus admin)
- AI:
 - GPT4+ (experimental design, software architecture); Genie.AI, Meshy.ai (text-to-3D)